

Old wine in new bottles: vitamin D in the treatment and prevention of tuberculosis.

Tuberculosis (TB) is a major cause of mortality, responsible for 1.68 million deaths worldwide in 2009. The global prevalence of latent *Mycobacterium tuberculosis* infection is estimated to be 32%, and this carries a 5-20% lifetime risk of reactivation disease. The emergence of drug-resistant organisms necessitates the development of new agents to enhance the response to antimicrobial therapy for active TB. Vitamin D was used to treat TB in the pre-antibiotic era, and its active metabolite, 1,25-dihydroxyvitamin D, has long been known to enhance the immune response to mycobacteria in vitro. Vitamin D deficiency is common in patients with active TB, and several clinical trials have evaluated the role of adjunctive vitamin D supplementation in its treatment. Results of these studies are conflicting, reflecting variation between studies in baseline vitamin D status of participants, dosing regimens and outcome measures. Vitamin D deficiency is also recognised to be highly prevalent among people with latent *M. tuberculosis* infection in both high- and low-burden settings, and there is a wealth of observational epidemiological evidence linking vitamin D deficiency with increased risk of reactivation disease. Randomised controlled trials of vitamin D supplementation for the prevention of active TB have yet to be performed, however. The conduct of such trials is a research priority, given the safety and low cost of vitamin D supplementation, and the potentially huge public health consequences of positive results.